SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I S.SACHIN/192525045 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sachin.S

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer

inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT enabled healthcare network. The Presentation Layer compresses the data by **30%,** the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightag e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understand in g of Concepts	25%	Demonstrat es a thorough, accurate understan din g of concepts	Good understandi ng with minor gaps	Basic understandi ng with some misconcepti on s	Poor understandin g; major misconcepti on s
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis

Organization	15%	Well structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1. File Transfer through Transport and Data Link Layer

Given

File Size = 150 MB

= 150 × 1024 × 1024

= **157,286,400 bytes**

Segment Size = **1500 bytes**

Frame Overhead = **40 bytes**

Bandwidth = **50 Mbps**

= 50×10^6 bits/s

Calculation

Number of Segments

= 157,286,400 ÷ 1500

= **104,858 segments**

Number of Frames

= **104,858 frames**

Total Overhead

= 104,858 × 40

= **4,194,320 bytes**

Total Data Transmitted

= 157,286,400 + 4,194,320

= **161,480,720 bytes**

Convert to bits

= 161,480,720 × 8

= **1,291,845,760 bits**

Transmission Time

= 1,291,845,760 ÷ 50,000,000

= **25.84 s**

Answer

Segments = **104,858**

Frames = **104,858**

Total Overhead = **4,194,320 bytes**

Transmission Time = **25.84 s**

2. Sliding Window Protocol

Given

Window Size = **6 frames**

Total Frames = **48**

Frame Size = **1024 bytes**

Calculation

Number of Windows

= 48 ÷ 6

= **8 windows**

Retransmissions

= 1 × 8

= **8 frames**

Answer

Transmission Windows = **8**

Retransmissions = **8 frames**

3. IP Fragmentation

Given

Packet Size = **12,000 bytes**

MTU = **1500 bytes**

IP Header = **20 bytes**

Calculation

Maximum Data per Fragment

= 1500 − 20

= **1480 bytes**

Number of Fragments

= 12,000 ÷ 1480

= 8.108

= **9 fragments**

Header Overhead

= 9 × 20

= **180 bytes**

Answer

Fragments = **9**

Header Overhead = **180 bytes**

4. Sliding Window Protocol

Given

Window Size = **6**

Frames = **48**

Calculation

Windows

= 48 ÷ 6

= **8**

Retransmissions

= 8 × 1

= **8**

Answer

Windows = **8**

Retransmissions = **8 frames**

5. Session Layer Synchronization

Given

File Size = **600 MB**

Checkpoint Interval = **60 MB**

Failure = **390 MB**

Calculation

Checkpoints Before Failure

= 390 ÷ 60

= 6.5

= **6 checkpoints**

Last Checkpoint

= 6 × 60

= **360 MB**

Retransmission

= 390 − 360

= **30 MB**

Answer

Checkpoints = **6**

Retransmitted Data = **30 MB**

6. Presentation Layer Compression

Given

Original File = **900 MB**

Compression = **40%**

Encryption Overhead = **5%**

Calculation

Compressed Size

= 900 × (100 − 40)/100

= 900 × 60/100

= **540 MB**

Encrypted Size

= 540 × 105/100

= **567 MB**

Reduction

= 900 − 567

= **333 MB**

Percentage Reduction

= (333 ÷ 900) × 100

= **37%**

Answer

Compressed Size = **540 MB**

Encrypted Size = **567 MB**

Reduction = **37%**

7. FTP Transfer

Given

Data = **3 GB**

= 3 × 1024

= **3072 MB**

Request Size = **3 MB**

Throughput = **120 Mbps**

Calculation

Requests

= 3072 ÷ 3

= **1024 requests**

Convert Data to bits

3 GB

= 3 × 1024 × 1024 × 1024 × 8

= **25,769,803,776 bits**

Transmission Time

= 25,769,803,776 ÷ 120,000,000

= **214.75 s**

Answer

Requests = **1024**

Transmission Time = **214.75 s**

8. End-to-End Delay

Given

Application = **4 ms**

Transport = **3 ms**

Network = **5 ms**

Data Link = **2 ms**

Physical = **1 ms**

Propagation = **18 ms**

Transmission = **12 ms**

Calculation

Total Delay

= 4 + 3 + 5 + 2 + 1 + 18 + 12

= **45 ms**

Answer

Total End-to-End Delay = **45 ms**

9. Frame Overhead

Given

Useful Data = **80 MB**

= 80 × 1024 × 1024

= **83,886,080 bytes**

Frame Size = **1600 bytes**

Overhead = **80 bytes**

Payload

= 1600 − 80

= **1520 bytes**

Calculation

Frames

= 83,886,080 ÷ 1520

= **55,188.21**

= **55,189 frames**

Total Overhead

= 55,189 × 80

= **4,415,120 bytes**

Efficiency

= (1520 ÷ 1600) × 100

= **95%**

Answer

Frames = **55,189**

Overhead = **4,415,120 bytes**

Efficiency = **95%**

10. Healthcare Network

Given

Original File = **240 MB**

Compression = **30%**

Segment Size = **1400 bytes**

Frame Overhead = **60 bytes**

Calculation

Compressed Size

= 240 × 70/100

= **168 MB**

Convert to bytes

= 168 × 1024 × 1024

= **176,160,768 bytes**

Segments

= 176,160,768 ÷ 1400

= **125,829.12**

= **125,830 segments**

Protocol Overhead

= 125,830 × 60

= **7,549,800 bytes**

Final Data

= 176,160,768 + 7,549,800

= **183,710,568 bytes**

Answer

Compressed File = **168 MB**

Segments = **125,830**

Protocol Overhead = **7,549,800 bytes**

Final Data Transmitted = **183,710,568 bytes**